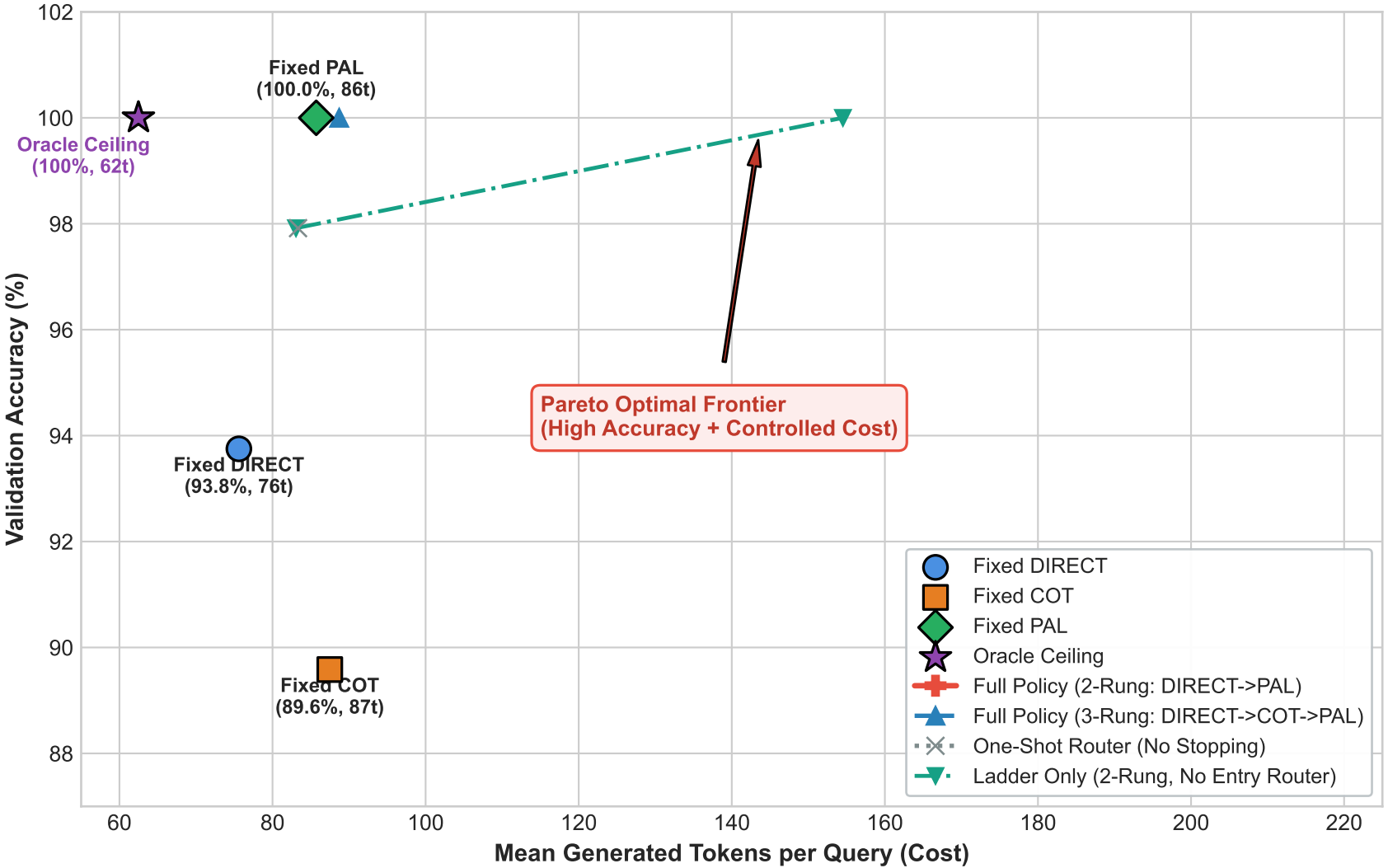
Pareto Frontier: Accuracy vs. Generated Token Cost on GSM8K (Qwen3-8B)
Validation Accuracy (%)
Mean Generated Tokens per Query (Cost)
102
100
98
96
94
92
90
88
60
80
100
120
140
160
180
200
220
Fixed PAL
(100.0%, 86t)
Oracle Ceiling
(100%, 62t)
Pareto Optimal Frontier
(High Accuracy + Controlled Cost)
Fixed DIRECT
(93.8%, 76t)
Fixed COT
(89.6%, 87t)
Fixed DIRECT
Fixed COT
Fixed PAL
Oracle Ceiling
Full Policy (2-Rung: DIRECT->PAL)
Full Policy (3-Rung: DIRECT->COT->PAL)
One-Shot Router (No Stopping)
Ladder Only (2-Rung, No Entry Router)